

EXPERIMENT NO. 4 — LINEAR REGRESSION

Learning Objective: Student should be able to implement Linear Regression.

Tools: Java / C / C++ / Python / Data Mining Tool

Theory: Implementation of Linear Regression

1. Linear Regression

Linear Regression is a supervised machine learning and statistical technique used to model the relationship between a dependent variable and one or more independent variables. In simple linear regression, there is one independent variable X and one dependent variable Y .

The objective is to find the straight line that best represents the relationship between the observed values of X and Y . Once the line is obtained, it can be used to estimate or predict Y for a new value of X .

$$Y = \alpha + \beta X$$

Y = predicted / dependent variable

X = independent variable

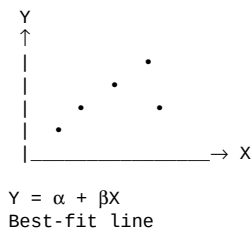
α = intercept

β = slope / regression coefficient

The **intercept** α represents the predicted value of Y when X is zero. The **slope** β represents the expected change in Y for a one-unit increase in X . If β is positive, Y tends to increase as X increases. If β is negative, Y tends to decrease.

2. Simple Linear Regression

Simple linear regression uses one independent variable to predict one dependent variable. The observations are represented as points on an X - Y graph, and regression finds the line that provides the best fit to those points.



The observed data points will usually not lie exactly on the regression line. The vertical difference between an observed value and its predicted value is called the **residual** or error.

$$\text{Residual} = \text{Actual } Y - \text{Predicted } Y$$

3. Least Squares Method

The most common method for estimating the regression coefficients is the **Ordinary Least Squares (OLS)** method. It selects the line for which the sum of the squared residuals is as small as possible.

Minimize:

$$\sum (Y_i - \hat{Y}_i)^2$$

where:

Y_i = actual value

$\hat{Y}_i$ = predicted value

Squaring the errors prevents positive and negative errors from cancelling each other and gives greater importance to larger errors.

4. Regression Coefficients

The slope and intercept can be calculated directly from the training data.

$$\beta = \frac{\sum [(X_i - \bar{X})(Y_i - \bar{Y})]}{\sum [(X_i - \bar{X})^2]}$$

$$\alpha = \bar{Y} - \beta \bar{X}$$

Therefore:

$$\hat{Y} = \alpha + \beta X$$

Here $\bar{X}$ is the mean of all X values and $\bar{Y}$ is the mean of all Y values. After calculating α and β , the model is ready to predict new values.

5. Positive and Negative Relationships

If the slope β is **positive**, the regression line rises from left to right. This indicates a positive relationship between X and Y. If β is **negative**, the line falls from left to right, indicating a negative relationship.

Slope	Relationship	Interpretation
$\beta > 0$	Positive	Y generally increases when X increases.
$\beta < 0$	Negative	Y generally decreases when X increases.
$\beta = 0$	No linear relationship	X does not explain a linear change in Y.

LINEAR REGRESSION — THEORY

6. Regression Line and Prediction

Once α and β have been calculated, the regression equation is used for prediction. For a new input $X = x$, the predicted output is obtained by substituting x into the equation.

Given:

$$\begin{aligned}\alpha &= 50 \\ \beta &= 30\end{aligned}$$

For $X = 8$:

$$\begin{aligned}Y &= 50 + (30 \times 8) \\ &= 290\end{aligned}$$

The predicted value is an estimate based on the relationship learned from the training data. It is not necessarily equal to the actual future value.

7. Correlation vs Regression

Correlation measures the strength and direction of association between two variables, whereas regression builds an equation that can be used for prediction. Regression explicitly defines independent and dependent variables.

Correlation	Regression
Measures association	Models relationship
Summarized by correlation coefficient r	Produces regression equation
Does not by itself define a prediction equation	Used for prediction
Symmetric relationship	Defines X and Y roles

8. Goodness of Fit — R^2

R^2 (coefficient of determination) indicates how much of the variation in the dependent variable is explained by the regression model.

$$R^2 = 1 - \text{SSE} / \text{SST}$$

$$\begin{aligned}\text{SSE} &= \sum (Y_i - \hat{Y}_i)^2 \\ \text{SST} &= \sum (Y_i - \bar{Y})^2\end{aligned}$$

For a standard model with an intercept, R^2 is commonly between 0 and 1. A value closer to 1 indicates that the fitted line explains a larger proportion of the observed variation. A high R^2 alone does not prove causation or guarantee that a model is appropriate.

9. Assumptions and Limitations

- The relationship between X and the mean of Y should be approximately linear.
- Extreme outliers can strongly influence the fitted line.
- Predictions outside the range of observed X values are extrapolations and can be unreliable.
- For statistical inference, residual behavior and other assumptions should also be examined.
- Simple linear regression models one predictor; multiple linear regression can use several predictors.

10. Applications

- Predicting salary from years of experience.
- Predicting sales from advertising expenditure.
- Estimating house prices from area.
- Forecasting demand using historical variables.
- Predicting marks from study hours.
- Estimating fuel consumption from engine-related measurements.

11. Overall Process

Training Data (X , Y)
↓

Calculate X and Y

↓

Calculate Slope β

↓

Calculate Intercept α

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Build model: $Y = \alpha + \beta X$

↓

Predict new Y values

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Compare Actual vs Predicted

PART B — SAMPLE EXAMPLE

Consider a simple example where **X = years of experience** and **Y = salary in thousands**, using the same type of data shown in the college experiment.

Experience X (Years)	Salary Y (■000)
2.5	55
4.0	225
6.5	300
11.0	350
12.5	475

12. Step 1 — Calculate Means

$$\bar{X} = (2.5 + 4 + 6.5 + 11 + 12.5) / 5 \\ = 7.3$$

$$\bar{Y} = (55 + 225 + 300 + 350 + 475) / 5 \\ = 281$$

13. Step 2 — Calculate Slope β

$$\beta = \frac{\sum[(X_i - \bar{X})(Y_i - \bar{Y})]}{\sum[(X_i - \bar{X})^2]}$$

$$\frac{\sum[(X_i - \bar{X})(Y_i - \bar{Y})]}{\sum[(X_i - \bar{X})^2]} = \frac{2518.5}{75.3}$$

$$\beta = 2518.5 / 75.3 \\ \beta \approx 33.4462$$

14. Step 3 — Calculate Intercept α

$$\alpha = \bar{Y} - \beta\bar{X}$$

$$\alpha = 281 - (33.4462 \times 7.3)$$

$$\alpha \approx 36.8426$$

Therefore, the fitted regression equation is:

$$\hat{Y} = 36.8426 + 33.4462X$$

15. Step 4 — Predict Salary

For a person with **8 years of experience**:

$$X = 8$$

$$\hat{Y} = 36.8426 + (33.4462 \times 8) \\ \hat{Y} \approx 304.4123$$

Predicted salary $\approx$ ■304.41 thousand

16. Original vs Predicted Values

Experience	Actual Salary	Predicted Salary (■000)
2.5	55	120.46
4.0	225	170.63
6.5	300	254.24
11.0	350	404.75
12.5	475	454.92

17. Interpretation

The positive slope of approximately **33.446** means that, according to this fitted model, one additional year of experience is associated with an estimated increase of about **■33.446 thousand** in predicted salary, within the range represented by the data. This is an association from the fitted model and should not automatically be interpreted as proof of causation.

18. Result and Conclusion

The Linear Regression algorithm was studied and implemented. The relationship between years of experience and salary was represented using the equation $\hat{Y} = \alpha + \beta X$. The slope and intercept were calculated using the least-squares method, and the model was used to predict salary for a new input value.

Linear regression is useful when a reasonably linear relationship exists between variables and a numerical output needs to be estimated.